

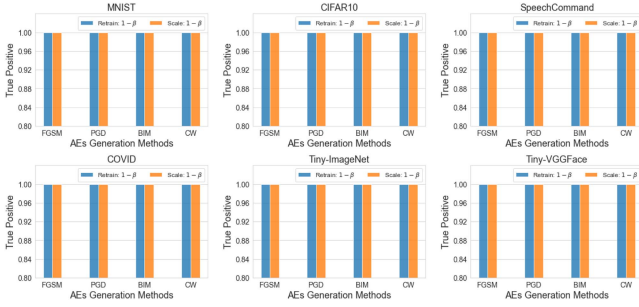


Fig. 10. The verified unlearning confidence through A2E watermark with different AEs generation methods.

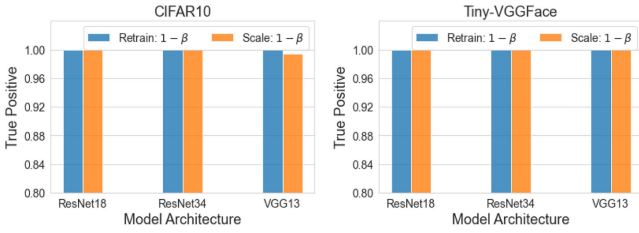


Fig. 11. The influence of model architecture on the verified unlearning confidence through A2E watermark on CIFAR10 and Tiny-VGGFace.

degree of uniqueness associated with the leaver's watermarking AEs. Put simply, $similarity \neq 1$, it implies that colluders are unable to generate such watermarking AEs that perfectly match the leaver and render the model completely immune to the leaver's watermarking AEs. Consequently, even enhanced with the colluders' watermarking AEs, the unlearned global model still outputs a lower P_0 and the following higher *true positive*, owing to it cannot achieve complete resistance to the leaver's distinctive watermarking AEs.

Remark 3: A2E watermark is reasonably robust to multiple adaptive strategies with $TP > 0.95$.

A. RQ3: Resilience to More Configurations

In this section, we further evaluate the resilience of A2E watermark against multiple configurations.

AEs generation methods: Fig. 10 explores the influence of different AEs generation methods on the verified unlearning confidence through A2E watermark. Overall, we apply four AEs generation methods, including untargeted FGSM, PGD, BIM and targeted CW. In general, different AEs generation methods have negligible impact on the unlearning verification effect of A2E watermark. Thus, choosing efficient FGSM, involved in A2E watermark generation, is sufficient to establish the confident unlearning verification effect.

Model architecture: Here, we use ResNet18 with fewer layers, ResNet34 with more layers and VGG13 with more parameters on CIFAR10 and Tiny-VGGFace. As Fig. 11 shows, A2E watermark can always demonstrate the highly confident verified unlearning effect across diverse models.

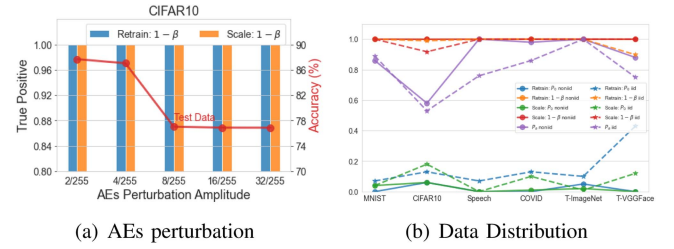


Fig. 12. The influence of perturbation amplitude and data distribution through A2E watermark.

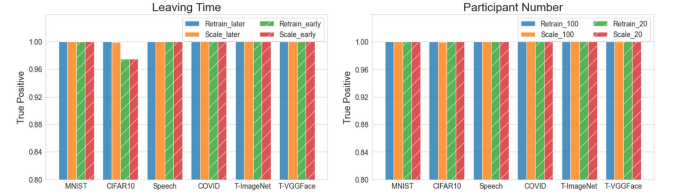


Fig. 13. The influence of leaving time (early or later) and participant number (more or less) on the verified unlearning confidence through A2E watermark.

AEs perturbation: Fig. 12(a) demonstrates the perturbation amplitude, used in the watermark-dependent AEs generation process, would hardly influence the verified unlearning result through A2E watermark on CIFAR10. However, as the perturbation amplitude increases, a more pronounced negative impact on the original learning task becomes evident, as indicated by a greater decline in accuracy on the test data. Consequently, while ensuring the confident unlearning verified, a lower perturbation amplitude is preferable to mitigate the adverse effects on the original learning task.

Data distribution (non-i.i.d. and i.i.d.): We further explore the influence of data distribution (non-i.i.d. and i.i.d.) on the unlearning verification result through our proposed A2E watermark. As shown in Fig. 12(b), i.i.d. setting has a smaller P_a and larger P_0 than non-i.i.d. setting. This matches the expectation as both watermarking and unlearning would be less effective in i.i.d. setting due to the more similar contributions of other non-leaving participants. On the other hand, the leavers tend to preserve more distinctive watermarking AEs for producing more unique traces under non-i.i.d. setting. As a result, after unlearning, P_0 is smaller on these more specialized watermark-dependent AEs under non-i.i.d. setting. Nevertheless, even under i.i.d. setting, A2E still achieves a highly confident unlearning verification effect with $TP > 0.9$ on all datasets, which shows the stability of A2E watermark under different data distribution.

Leaving time: We also analyze the influence of leaving time on our approach. Specifically, we designate the early leaving time as 100 rounds previous to the later leaving time $t_{unlearn}$ in Table I. As Fig. 13 shows, the verified unlearning confidence through A2E watermark is similar between early and later leaving. Although a slight drop is observed on CIFAR10 during early leaving, *true positive* still exceeds 0.97. Overall, these findings suggest that the leaving time has a negligible impact on the verified unlearning confidence.